Ahmet Bilal Yazicioğlu

Istanbul, Turkey |
bilalyazicioglu.com | linkedin.com/in/bilal-yazicioglu | github.com/bilalyazicioglu

Summary

Computer Engineering student focused on systems, infrastructure, networking, and backend engineering. I build and operate software end to end, from **Rust/QUIC peer-to-peer systems** to self-hosted **Docker/Nginx** services with CI/CD and observability. Hands-on experience with open-source development, backend systems, Linux, networking, and debugging real-world software across multiple environments.

Education

Marmara University — Istanbul, TR
B.S. in Computer Engineering — Minor in Business Administration — *2024 – 2028 (Expected)*

Universidad de Oviedo — Oviedo, ES
Erasmus+ Exchange Student, Computer Science — *2026*

Selected Projects

tincan — P2P Voice & Text Chat (200+ GitHub Stars) *2026*
Rust, iroh/QUIC, Opus, ratatui • tincan.rs • GitHub

- Built and maintain an open-source terminal communication system with no central voice server, distributed through Homebrew, npm, Cargo, and prebuilt binaries.
- Designed a coordinator-based room protocol and direct peer audio mesh over QUIC datagrams with Opus audio, jitter buffering, Argon2id-based room authentication, relay fallback, and 93 automated/integration tests.
- Debugged peer connectivity, relay fallback, audio transport, cross-platform behavior, and failure handling across real-world network conditions.

Personal Portfolio & Monitoring Stack *2026*
Next.js, TypeScript, Docker, Nginx, Cloudflare, Tailscale, Prometheus/Loki/Grafana • bilalyazicioglu.com

- Self-host and operate a containerized web stack behind Nginx and Cloudflare Tunnel, with private Tailscale access, CI/CD, and Prometheus/Loki/Grafana for metrics, logs, and operational monitoring.

Independent AR & 3D Reconstruction Work *2026*
Node.js, Redis, Python, Swift, Three.js, Docker, GitHub Actions

- Experimenting with augmented reality and multi-image 3D reconstruction workflows, focusing on backend processing pipelines, system integration, automated testing, deployment, and observability.

Experience

Kombi Klima Shop — Istanbul, TR
Software Developer Intern — *2025*

- Extended an e-commerce backend in **Java/Spring Boot**, improving system performance and order-processing workflows.
- Built a **SQL**-based stock automation system that replaced manual inventory tracking and reduced counting errors.

Software Architecture Project (yovi_en1b) — Oviedo, ES
Developer / Architect — Universidad de Oviedo — *2026*

- Built a multi-service hexagonal game engine in **Rust** in a six-person international team, owning game logic and bot development.
- Worked with Docker deployment, GitHub Actions CI/CD, automated testing, and arc42 architecture documentation.

Open Source Contributions

instagram-cli (PR #388) Open Source
Open Source Contribution *2026*

- Implemented drag-and-drop file handling for terminal input, normalizing dropped absolute paths and integrating them with the existing file-upload preprocessing pipeline.
- Added automated coverage for paths containing spaces, quotes, and image inputs, along with diagnostic logging and documentation updates.

iroh (GitHub) Open Source
Ecosystem / Community Contribution *2026*

- Built Tincan on top of iroh's QUIC-based peer-to-peer networking stack and shared the project with the iroh community as a real-world application of the ecosystem.
- Investigated peer connectivity, relay behavior, connection resilience, and network-transition scenarios while integrating iroh into a real-time communication system.

Tuta Open Source
Localization Contributor *2026*

- Contributed localization work to Tuta, a privacy-focused open-source email and calendar platform.

Skills

Languages: Rust, Go, Java, Python, C, SQL, TypeScript, JavaScript, Bash

Systems & Networking: Linux, TCP/IP, DNS, HTTP/TLS, QUIC, P2P Networking, SSH

Infrastructure & Observability: Docker, GitHub Actions, Jenkins (basic), Nginx, Cloudflare, Tailscale, Prometheus, Grafana, Loki

Backend & Data: Spring Boot, Node.js/Express, REST APIs, PostgreSQL, MySQL, MongoDB, Redis

Web: React, Next.js, Three.js **Spoken**: Turkish (Native), English (Proficient), Spanish (Elementary)

Selected Coursework

Deep Learning Specialization — Andrew Ng, DeepLearning.AI: Neural networks, CNNs, sequence models, optimization, hyperparameter tuning, and practical deep learning with Python/TensorFlow.

Go | Modern Programming from Scratch — Udemy: Go fundamentals, REST APIs, concurrency, interfaces, error handling, unit/integration testing, and Docker.

Leadership & Activities

Proactivanet Hackathon — 4th Place / 12 Teams *2026*
Universidad de Oviedo

- Placed 4th among 12 teams in a two-day hackathon, developing an AI-assisted approach to identify and predict risks in physical backup processes for a real Proactivanet case.
- Worked in a multinational team of students from Spain, Romania, Czechia, and Turkey, collaborating across English and Spanish to deliver and present an MVP.

University Basketball *10+ Years*
Marmara University and Universidad de Oviedo

- Played competitive basketball for more than 10 years, including university teams at Marmara University and Universidad de Oviedo, developing teamwork, communication, discipline, and decision-making under pressure.
- Competed at Universidad de Oviedo in a multilingual and international team environment during Erasmus+.

ESN Oviedo *2026*
Erasmus Student Network

- Participated in ESN Oviedo and international student activities during Erasmus+, engaging with students from diverse cultural and academic backgrounds.